Pranav Bhalerao

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Education

Vishwakarma Institute of Technology- Electronics and Telecommunication Engineering GPA 8.3 (SEM I- IV)

Skills :

Languages: C++, Java, Python

Developer Tools: VS code, Git/GitHub

Embedded Systems: IoT Development, ESP8266, Microcontrollers

Projects

Weather App – Python

Python, OpenWeatherMap API, Tkinter (GUI), Requests, JSON

• Developed a GUI-based weather application using Python to fetch and display real-time weather data for user-searched cities. Integrated the OpenWeatherMap API to retrieve accurate weather conditions including temperature, humidity, and overall climate status.

• Implemented city-based search functionality with validation to ensure correct input of city names.

• Designed a dynamic user interface using Tkinter, providing an interactive and user-friendly experience.

• Enhanced UI/UX by updating background images dynamically based on weather conditions (e.g., sunny, rainy, cloudy).

• Displayed key environmental parameters such as temperature, humidity, and weather descriptions (e.g., Sunny, Rainy, Cloudy) in real time.

LoRa-Based Environmental Monitoring System
ESP8266, LoRa Trans/Receiver SX1278, Sensors, IoT

- **Developed a long-range wireless environmental monitoring system using LoRa communication.**
- **Collected sensor data such as temperature and humidity from remote locations and transmitted it between LoRa transmitter and receiver modules.**
- **Integrated microcontrollers to process and relay sensor data for monitoring applications.**
- **Designed the system for low-power, long-distance data transmission suitable for agricultural and remote field monitoring.**

Fire Fighting Robot
Arduino, Flame Sensor, DC Motors, Motor Driver

- **Built an autonomous robot capable of detecting fire using flame sensors.**
- **Implemented motor control logic for navigation toward the fire source.**
- **Integrated a water pump mechanism to extinguish detected flames.**
- **Designed and tested embedded control logic for real-time fire response.**

Certification

International Conference Paper Presentation – ICEAMST 2025
RV College of Engineering, Bengaluru | IEEE Electron Devices Society

- **Presented research paper titled "An Affordable Autonomous Firefighting Robot for Residential and Small-Scale Settings."**
- **Demonstrated the design and implementation of a sensor-based firefighting robotic system.**
- **Showcased autonomous fire detection and suppression using embedded systems and robotics.**

Achievements

- Presented a research paper titled *"An Affordable Autonomous Firefighting Robot for Residential and Small-Scale Settings"* at the **International Conference on Emerging Applications of Material Science and Technology (ICEAMST 2025)**.